# Case Study
# as an
# Engineering Research Method

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# 1 When to consider this research method

## 1.1 Humanities vs engineering research

In humanities research, the differences between quantitative and qualitative research are emphasised due to the nature of research in that area where it is usually infeasible to create experiments that are independently repeatable and that produce generally proven results. Some research in engineering contexts, such as studying user experiences and public perceptions in engineering projects, should fully adopt the methodologies of humanities research. The guidelines presented below are not aimed at such scenarios. A useful reference for more detailed information about case study research is that of Hancock and Algozzine [2006]. It is aimed at explaining the methodology to postgraduate students, but is situated in educational research.

Most engineering research projects primarily use quantitative research methods, e.g. numerical analysis, laboratory experiments or, in some respects, design. Such projects, however, often also involve qualitative methods, such as case studies. Strictly speaking, these projects then use a mixed methodology.

## 1.2 Research planning context

Consider the typical steps involved in research planning (adapted from Hancock and Algozzine [2006]):

1. Determining what we want to study (the research question)
2. Determining how we want to study it (the methodology)
3. Determining what application we want to study (the case, cases or sample)
4. Determining how best to acquire information (the data-collection techniques)
5. Determining how best to analyse or interpret the information that we acquire (the data analysis)
6. Determining how to confirm our findings (the verification process)
7. Determining how and with whom to share our findings (the dissemination process)

The answers in Steps 3 and 4 above will determine whether a "case study" is conducted.

# 2 Important aspects in a case study

## 2.1 Purpose

A case study method is used when the researchers select one or a small number of cases to study in depth, instead of attempting to prove more general claims by testing a hypothesis on an exhaustive range of scenarios. In engineering research, the term "case study" is usually applied to implementation scenarios

that are close to "real-world" applications. The inherent intention in case study research is to focus in detail on the problem in a particular context, acknowledging that the research results may have implications for other contexts, but not trying to prove the broader implications. However, limiting the scope of the project to the selected cases, allows the researchers to explore those cases in greater detail.

## 2.2 Context

A case study investigates a situation/application/example bounded by space and time, ideally in its natural context. In engineering research, it is sometimes not feasible to use the case's natural context, and then a simulated or simplified context is used. A case typically can be a subset of the intended end-users, expected events or intended applications. Although case studies are situated at a specific instance in time, the results in engineering research are often independent of the instance in time. Researchers formulating case studies should, however, consider this aspect.

A case study gives the researcher a manageable context so that the case can be investigated in detail in a natural context (preferably) or a simulated context. The research can use all of the case's properties and context as sources of information, thereby studying complex interactions in detail.

## 2.3 Generality of the research results

The richness of information in a case study comes at a price: the research results will only be proven for the chosen case. The researcher will have to argue or speculate to what extent the case study's results are applicable in other cases.

There are, however, four methods that can be used to broaden the claims of a case study's results (Greene, Caracelli and Graham, 1989, in Onwuegbuzie and Leech, 2005, p. 384):

- **Triangulation:** the purpose of triangulation is to seek ‘convergence and corroboration’ (ibid.) via different methods. For example, the researcher would attempt to get the same results using different PLCs, and detail the differences.
- **Complementarity:** this is the process of enhancing, illustrating and clarifying ‘results from one method with the results from the other method’ (ibid.).
- **Development:** the use of the results of method A to inform method B. Here a researcher could use an existing solution from a similar case (in the literature or in the current context) and apply it in the case study context.
- **Expansion:** the need ‘to expand the breadth and range of inquiry by using different methods’ (ibid.). For example, using a solution from a different kind of problem scenario that has not been used in contexts such as the current the case study.

# 3 Examples

## 3.1 A software architecture

Consider a research project developing a new manufacturing execution control architecture that is aimed at being widely applicable. The architecture can best be evaluated by implementing and testing the implementation. However, the research project will typically only implement the architecture using one or two particular implementation platforms (e.g. agent-based programming or service-oriented architectures), while other implementation platforms could have been used.

The research project will, further, typically test the implementation in one or two laboratory or industrial case studies. The laboratory cases will be simulations or idealisations of real-life industrial cases. For the chosen cases, however, detailed observations and data collection will be possible.

The evaluation of the architecture should then be considered to be a case study, with the advantages that the feasibility of the architecture could be proved and that detailed, quantitative results could be obtained. The researchers may give arguments why they consider the architecture to be more widely applicable, but the claim is only verified for situations similar to the case study.

## 3.2 A machine design

If the research project is aimed at developing a machine that can be used in a wide range of contexts, it will typically only be possible to test the machine in a few limited contexts. The testing will then employ a case study method and the selected contexts constitute the selected cases. Detailed results can be obtained for the selected cases and the performance of the machine can be assessed against quantitative targets.

As for the other examples, the researchers may give reasons why they consider the machine to be widely applicable, but the research report should acknowledge that the claim is unverified.

## 3.3 Techno-economic analyses

Consider a research project aimed at developing a technic-economic model of a broad application, such as gravity-based energy storage. Because the model is intended for broad application, it cannot be evaluated for all possible applications. Therefore, a case study method is more appropriate, in which the model is evaluated for a finite set of selected scenarios.

As for the other examples, the researchers may argue that the model is more widely applicable, but have to acknowledge that the application has only been proven for the selected cases.

# 4 Conclusion

In a research proposal or report for a project that uses a case study, it is important to:

1. Defend the use of a case study method, with due consideration of the strengths and limitations of the method, as outlined in Sections 2.1 and 2.2.
2. Motivate the selection of the cases, with due consideration of the strengths and limitations of the selected cases. A review of the literature or the needs in the industry (e.g. common or new needs) could be used to support a choice of case study.
3. Restrict claims based on the case study results to the case study's context, as outlined in Section 2.3.
4. Argue (or speculate) to what extent the results could be more widely applicable, but do not claim such wider application without evidence.

# References

Hancock, DR and Algozzine, B: *Doing Case Study Research, A Practical Guide for Beginning Researchers*, Teachers College Press, New York, 2006

Onwuegbuzie, AJ and Leech, NL: On becoming a pragmatic researcher: The importance of combining quantitative and qualitative research methodologies. *International Journal of Social Research Methodology*, 8 (5) 2005: 375-387